Unit 2 Guided Notes — Scientific Notation

ANSWER KEY

Big or tiny number? Pack it into a × 10^{n}.

🎯 By the end of this unit, I can…

- Write and compute with numbers in scientific notation

Standards: NC.8.EE.3 · NC.8.EE.4

Answer key: every blank and final answer is filled in green.

1 Scientific Notation

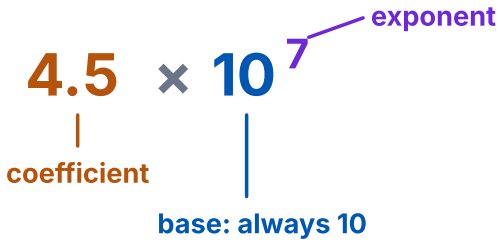
📖 Words to know

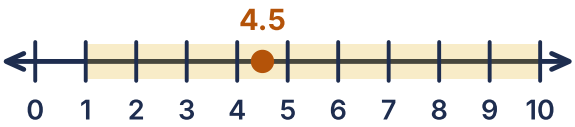
Standard form: a number written out with all its digits, like 45,000,000.

Scientific notation: a number written as (a number from 1 up to 10) × 10 to a power, like 4.5 × 10^{7}.

Coefficient: the first number in scientific notation. In 4.5 × 10^{7}, the coefficient is 4.5.

📌 **THE FORMAT:** A number in scientific notation looks like 4.5 × 10^{7}. The first number must be at least 1 and less than 10. The base is always 10.

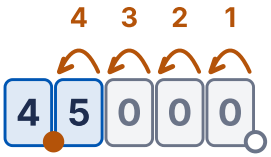




the coefficient must be from 1 up to (not including) 10: the shaded part

Move the Decimal

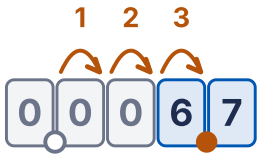
45,000 (big number)



$= 4.5 \times 10^{4}$

moved LEFT 4 → exponent +4

0.0067 (small number)



$= 6.7 \times 10^{-3}$

moved RIGHT 3 → exponent −3

The gold dot is where the decimal point lands: right after the first digit that is not 0. Each hop is one power of 10.

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	POSITIVE	45,000 = 4.5×10^{4}
Small number (< 1)	Move decimal RIGHT	NEGATIVE	0.0067 = 6.7×10^{-3}

Multiplying — 3-Step Process

1. MULTIPLY the coefficients (the a-values)
2. ADD the exponents
3. Adjust if coefficient is NOT between 1 and 10

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

$$(3 \times 10^4) \times (2 \times 10^5)$$

Regroup: numbers together, tens together

$$(3 \times 2) \times (10^4 \times 10^5)$$

3 × 2 = ? | add: 4 + 5 = ?

$$= ? \times 10^?$$

Multiply coefficients: 3 × 2 = 6

ADD exponents: 4 + 5 = 9

✓ **Answer:** 6×10^9

Dividing — 3-Step Process

1. **DIVIDE** the coefficients
2. **SUBTRACT** the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

$$\frac{8 \times 10^6}{4 \times 10^2} = \frac{8}{4} \times \frac{10^6}{10^2}$$

Split it: numbers ÷ numbers, tens ÷ tens

8 ÷ 4 = ? | subtract: 6 − 2 = ?

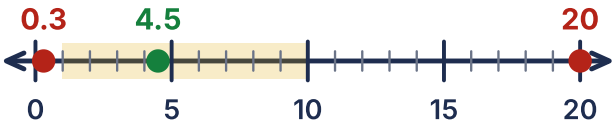
$$= ? \times 10^?$$

Divide coefficients: 8 ÷ 4 = 2

SUBTRACT exponents: 6 − 2 = 4

✓ **Answer:** 2×10^4

⚠️ **ADJUSTMENT RULES:** If coefficient ≥ 10: move decimal LEFT, ADD 1 to exponent. If coefficient < 1: move decimal RIGHT, SUBTRACT 1 from exponent.



Only coefficients in the gold band (1 up to, not including, 10) are finished. 20 and 0.3 still need one more hop.

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

$$(5 \times 10^3) \times (4 \times 10^2)$$

Regroup: numbers together, tens together

$$(5 \times 4) \times (10^3 \times 10^2)$$

5 × 4 = ? | add: 3 + 2 = ?

$$= ? \times 10^?$$

Last: is the coefficient from 1 up to 10?

Multiply: 5 × 4 = 20

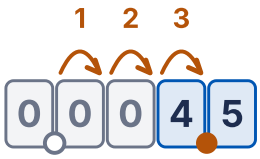
Add exponents: 3 + 2 = 5 → result: 20×10^5

Adjust (20 ≥ 10): $20 = 2.0 \times 10^1$, so ADD 1 to the exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Convert 0.0045 to scientific notation. A student wrote 4.5×10^3. What's the mistake?**

$$0.0045$$



$$= 4.5 \times 10^?$$

0.0045 is less than 1, and the point hops to the RIGHT. Which sign must the exponent have?

Small number → NEGATIVE exponent. Correct: 4.5×10^{-3}. Numbers less than 1 always have negative exponents in scientific notation.

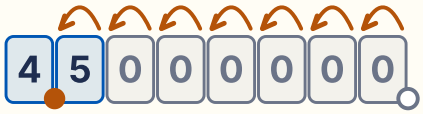
Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

A. Scientific Notation Conversions

1) 45,000,000 → scientific notation

45,000,000



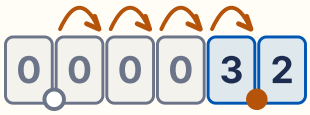
= ? × $10^{?}$

Count the hops to the gold dot.

Answer: 4.5×10^{7}

2) 0.00032 → scientific notation

0.00032



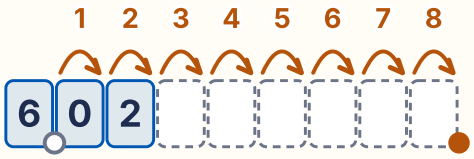
= ? × $10^{?}$

Count the hops. Which way did the point move?

Answer: 3.2×10^{-4}

3) 6.02×10^{8} → standard form

6.02×10^{8}



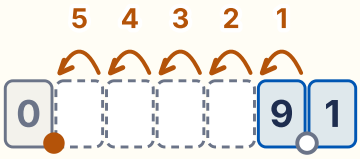
= ?

Hop 8 places right. Fill each empty box with 0.

Answer: 602,000,000

4) 9.1×10^{-5} → standard form

9.1×10^{-5}



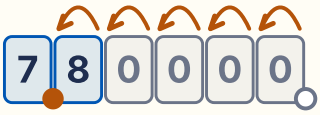
= ?

Hop 5 places left. Fill each empty box with 0.

Answer: 0.000091

5) 780,000 → scientific notation

780,000



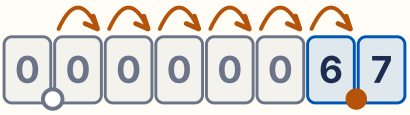
= ? × $10^{?}$

Count the hops to the gold dot.

Answer: 7.8×10^{5}

6) 0.0000067 → scientific notation

0.0000067



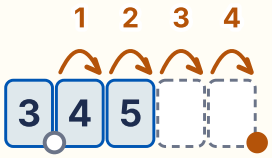
= ? × $10^{?}$

Count the hops. Which way did the point move?

Answer: 6.7×10^{-6}

7) 3.45 × 10^{4} → standard form

3.45 × 10^{4}



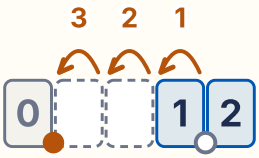
= ?

Hop 4 places right. Fill each empty box with 0.

Answer: 34,500

8) 1.2 × 10^{-3} → standard form

1.2 × 10^{-3}



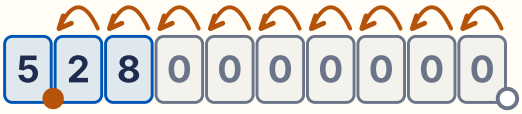
= ?

Hop 3 places left. Fill each empty box with 0.

Answer: 0.0012

9) 5,280,000,000 → scientific notation

5,280,000,000



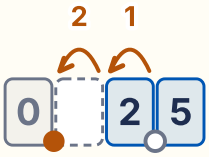
$= ? \times 10^{?}$

Count the hops to the gold dot.

Answer: 5.28×10^{9}

10) 2.5×10^{-2} → standard form

2.5×10^{-2}



$= ?$

Hop 2 places left. Fill each empty box with 0.

Answer: 0.025

B. Operations

1) $(3 \times 10^{4})(2 \times 10^{5})$

(3×10^{4}) 3 10 10 10 10

$\times (2 \times 10^{5})$ 2 10 10 10 10 10

$= ? \times 10^{?}$

Multiply the gold numbers. Count all the 10 tiles.

Answer: 6×10^{9}

2) $(8 \times 10^{7}) \div (4 \times 10^{3})$

8 10 10 10 10 10 10 10

4 10 10 10

$= ? \times 10^{?}$

Divide the gold numbers. Cancel matching 10s, top and bottom.

Answer: 2×10^{4}

3) $(5 \times 10^3)(7 \times 10^6)$

(5×10^3) 5 10 10 10

$\times (7 \times 10^6)$ 7 10 10 10 10 10 10

$= ? \times 10^?$

Multiply, count the 10s, then check: is the coefficient from 1 up to 10?

Answer: 3.5×10^{10}

4) $(9.6 \times 10^8) \div (3.2 \times 10^5)$

9.6 10 10 10 10 10 10 10 10

3.2 10 10 10 10 10

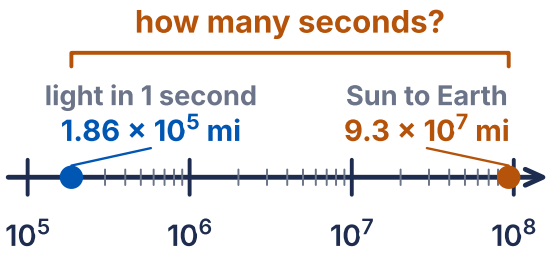
$= ? \times 10^?$

Divide the gold numbers. Cancel matching 10s, top and bottom.

Answer: 3×10^3

C. Word Problems

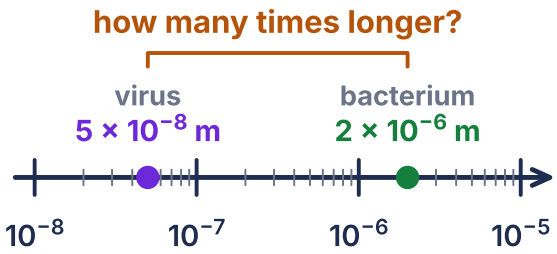
1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds does it take light to reach Earth?



Each tick is 10 times the one before. How many 1-second trips fit in the whole distance?

$(9.3 \times 10^7) \div (1.86 \times 10^5) = 5 \times 10^2$ seconds (500 seconds).

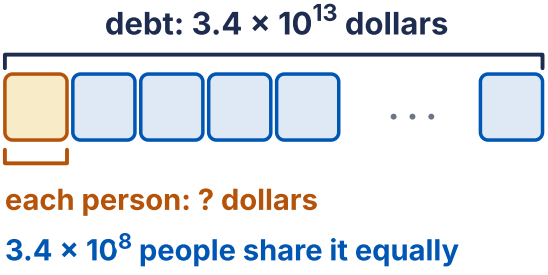
2. A bacterium is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacterium?



The bacterium is further right, so it is longer. Divide to find how many times.

$(2 \times 10^{-6}) \div (5 \times 10^{-8}) = 0.4 \times 10^{2} = 40$ times longer.

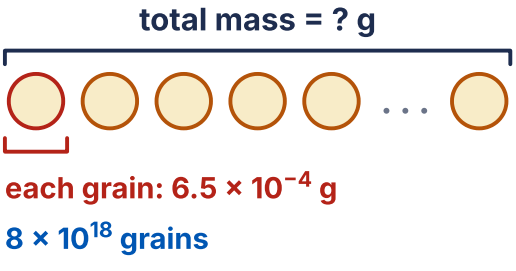
3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^{8} people. What is the debt per person?



Split the whole debt into one equal share per person.

$(3.4 \times 10^{13}) \div (3.4 \times 10^{8}) = 1 \times 10^{5} = \$100{,}000$ per person.

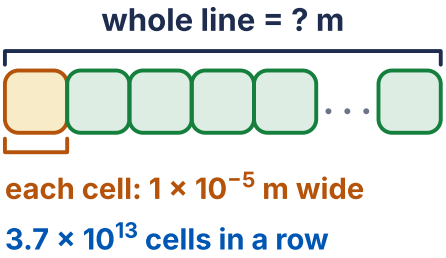
4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?



Same mass, many times: multiply.

$(6.5 \times 10^{-4})(8 \times 10^{18}) = 52 \times 10^{14} = 5.2 \times 10^{15}$ grams.

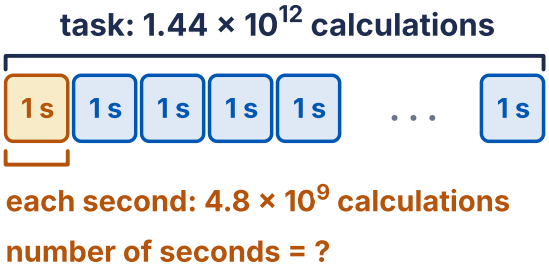
5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?



Same width, many times: multiply.

$(3.7 \times 10^{13})(1 \times 10^{-5}) = 3.7 \times 10^{8}$ meters.

6. A computer does 4.8×10^{9} calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?



Cut the whole task into 1-second pieces. How many pieces?

$(1.44 \times 10^{12}) \div (4.8 \times 10^{9}) = 0.3 \times 10^{3} = 3 \times 10^{2}$ seconds (300 seconds).